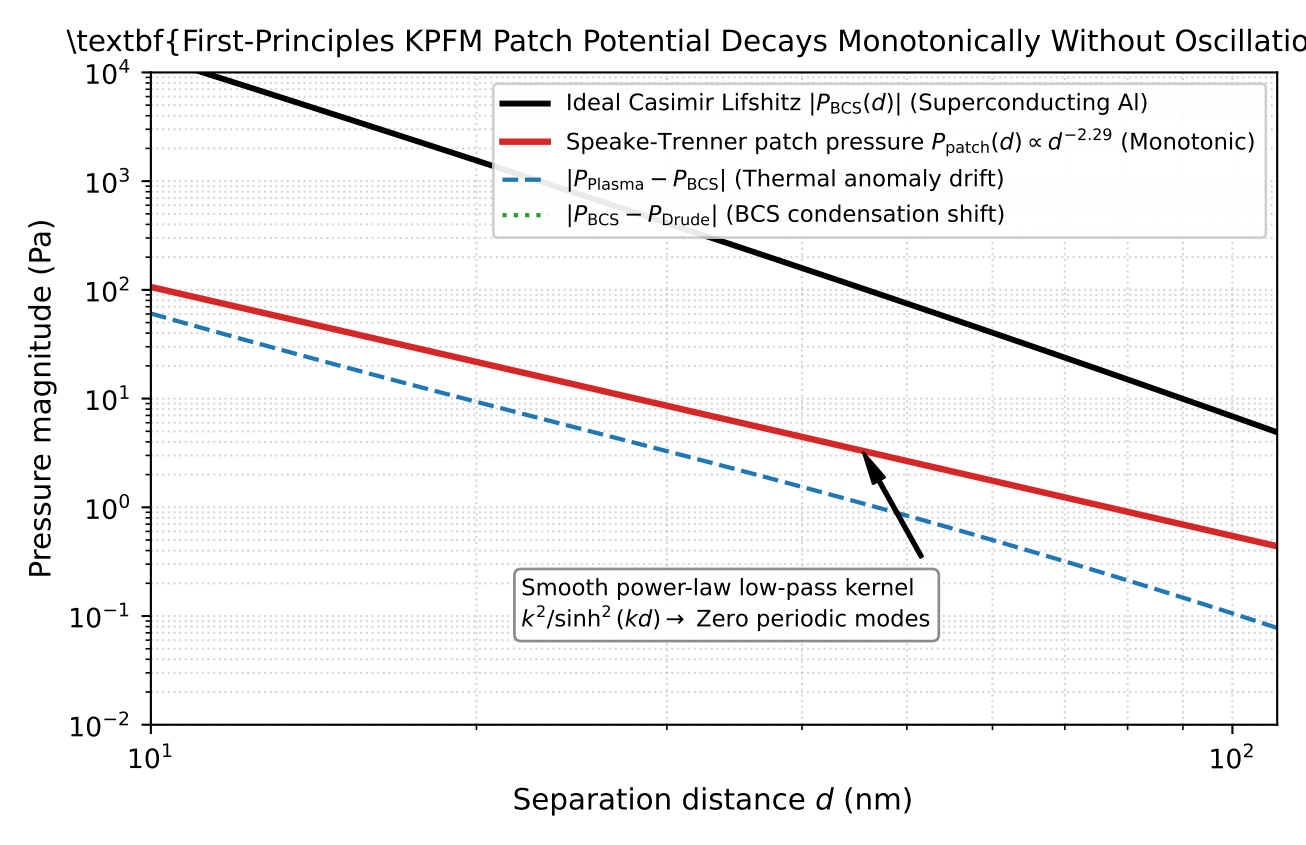
\textbf{First-Principles KPFM Patch Potential Decays Monotonically Without Oscillatio
Ideal Casimir Lifshitz $|P_{BCS}(d)|$ (Superconducting Al)
Speake-Trenner patch pressure $P_{patch}(d) \propto d^{-2.29}$ (Monotonic)
$|P_{Plasma} - P_{BCS}|$ (Thermal anomaly drift)
$|P_{BCS} - P_{Drude}|$ (BCS condensation shift)
Smooth power-law low-pass kernel
$k^2/\sinh^2(kd) \rightarrow$ Zero periodic modes
Pressure magnitude (Pa)
Separation distance d (nm)
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-2}
10^1
10^2